

CHECK WHAT YOU LEARNT

Check Yourself - Unit 2**A. Choose the correct answer**

1. `analogRead()` returns a number between
(a) 0 and 255 (b) 0 and 1023 (c) 0 and 5 (d) 0 and 100
2. `analogWrite()` accepts a number between
(a) 0 and 1023 (b) 0 and 255 (c) 0 and 10 (d) 0 and 5
3. `map(raw, 0, 1023, 0, 255)` is used to
(a) read a sensor (b) convert one range to another (c) print a value (d) add a delay
4. PWM works by
(a) lowering the voltage (b) switching on and off very fast (c) using a resistor (d) using the analog pins
5. Which pin can fade an output?
(a) D2 (b) D4 (c) D9 (d) D7
6. `Serial.begin(9600)` belongs inside
(a) `loop()` (b) `setup()` (c) `map()` (d) a comment

B. Fill in the blanks

- 1 A whole number is stored in a variable of type _____.
- 2 A number with a decimal point is stored in a _____.
- 3 PWM stands for Pulse _____ Modulation.
- 4 PWM-capable pins are marked with the symbol _____.

C. Answer in one or two lines

- 1 Explain why 1023 must be converted to 255 before `analogWrite()`.
- 2 Describe in your own words how a pin that is only ever on or off can dim an LED.
- 3 Your Serial Monitor shows strange symbols instead of numbers. What is wrong?
- 4 Why would storing a temperature of 27.4 in an `int` be a bad idea?

UNIT 3

Basic Project: LED Dimmer and Mood Light

Sessions 6, 7 and 8

Your first project of the year has no sensor at all - the sensor is you. A knob becomes an analog input, and three LEDs become an output that changes smoothly rather than jumping. Everything in Units 4 and 5 is built on this.

By the end of this unit you will be able to:

- Wire a potentiometer as a voltage divider and read it.
- Convert the reading into a brightness with `map()`.
- Control an LED smoothly using `analogWrite()`.
- Divide a range into zones and light a different lamp in each.

SESSION 6

one class

Reading the Potentiometer

Today you will

- Learn how a potentiometer turns rotation into voltage.
- Wire it correctly and read it on the Serial Monitor.
- Record the readings at both ends and in the middle.

A potentiometer - everyone calls it a **pot** - is a resistor with a slider. It has three legs. The two outer legs go to 5V and GND, so a fixed voltage sits across the whole strip. The middle leg, called the **wiper**, touches the strip at whatever point the knob has turned to. Turn the knob and the wiper picks up a different voltage, anywhere from 0V to 5V.

That is the whole trick, and it has a name: a **voltage divider**. It is how a volume knob works, how a joystick works, and how a car's accelerator pedal tells the engine how hard you are pressing.

Knob position	Voltage on the wiper	analogRead gives about
Turned fully anticlockwise	0 V	0
Halfway	2.5 V	512
Turned fully clockwise	5 V	1023

Watch Out

If your reading jumps straight from 0 to 1023 with nothing in between, you have almost certainly put the wiper on an outer leg. The middle leg is always the one that goes to A0.

Try This

Reverse the two outer legs - 5V and GND the other way round. Read the values again. The knob now counts backwards. Nothing is broken; you have simply flipped which end of the strip is which. Put it back before the next session.

SESSION 7

one class

Building the Dimmer

Today you will

- Wire the pot and three LEDs on the breadboard.
- Use map() and analogWrite() to control brightness.
- Get a smooth fade across the full turn of the knob.

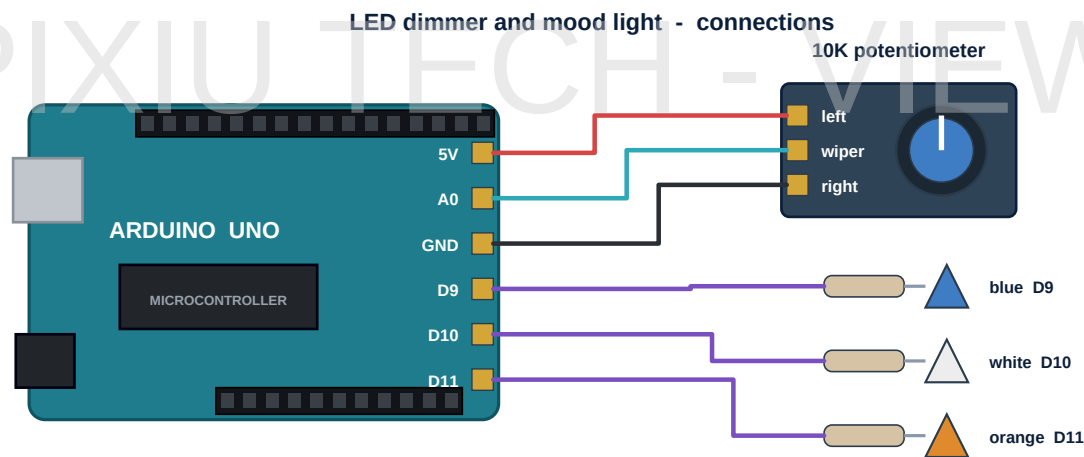


Figure 3.1 - One knob in, three lamps out

Component pin	Arduino pin	Why
Pot - left leg	5V	One end of the strip
Pot - right leg	GND	The other end
Pot - middle leg (wiper)	A0	The changing voltage you read
Blue LED long leg	D9 (~) through 220 ohm	PWM-capable pin
White LED long leg	D10 (~) through 220 ohm	PWM-capable pin
Orange LED long leg	D11 (~) through 220 ohm	PWM-capable pin
All LED short legs	GND	Shared return path

LedDimmer.ino

```

int potPin = A0;
int ledPin = 9;           // must be a pin marked with ~

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(potPin);           // 0 to 1023
  int bright = map(raw, 0, 1023, 0, 255); // 0 to 255

  analogWrite(ledPin, bright);           // fade the LED

  Serial.print("raw: ");
  Serial.print(raw);
  Serial.print("  brightness: ");
  Serial.println(bright);

  delay(50);
}

```

Notice `Serial.print` without the `\n` for the first three, and `Serial.println` only on the last one. That keeps all four pieces on a single line and starts a new line at the end. It is a small detail that makes your output readable instead of a mess.

The Mood Light Challenge

Today you will

- Divide the knob's range into three zones.
- Light a different LED in each zone.
- Test the boundaries and fix the flickering.

A dimmer controls *how much*. A mood light also decides *which*. Split the range from 0 to 1023 into three equal zones and give each zone its own colour. The knob now does two jobs at once: it chooses the lamp and sets the brightness.

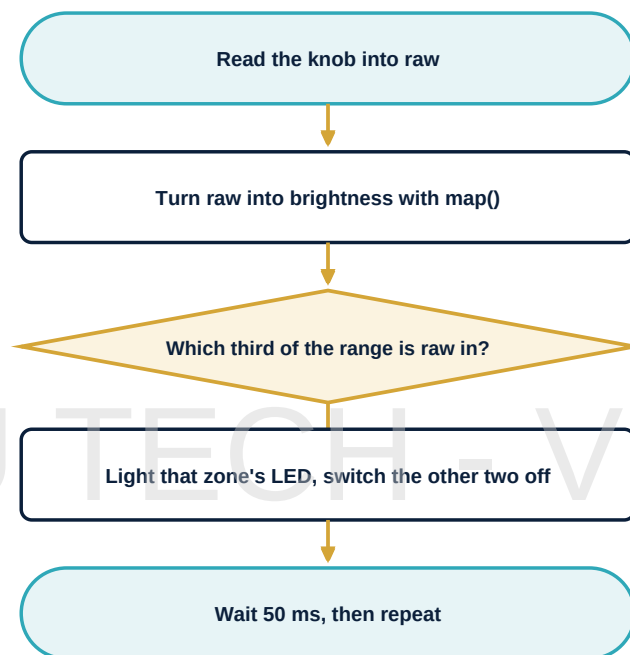


Figure 3.2 - One input choosing between three outputs

MoodLight.ino

```

int potPin = A0;
int blueLed = 9, whiteLed = 10, orangeLed = 11;

void setup() {
  pinMode(blueLed, OUTPUT);
  pinMode(whiteLed, OUTPUT);
  pinMode(orangeLed, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(potPin);
  int bright = map(raw, 0, 1023, 0, 255);

  analogWrite(blueLed, 0); // start with all three off
  analogWrite(whiteLed, 0);
  analogWrite(orangeLed, 0);

  if (raw < 341) {
    analogWrite(blueLed, bright); // calm zone
  } else if (raw < 682) {

```

```

    analogWrite(whiteLed, bright);    // work zone
  } else {
    analogWrite(orangeLed, bright);  // warm zone
  }

  delay(50);
}

```

Read the if - else if - else carefully. It is called a **decision ladder**, and the order matters enormously. The board checks the first condition; only if that is false does it try the second. Because of that, you never need to write *raw is between 341 and 682* - by the time the second rung is reached, the board already knows raw is at least 341.

Watch Out

Right at a boundary - around 341 or 682 - the reading wobbles by a few units and two LEDs appear to flicker against each other. That is not a fault in your code; it is a real engineering problem with a real name, and you will solve it properly in Unit 4.

Challenge

Add the buzzer to a PWM pin and use `analogWrite()` on it as well, so the knob changes light and sound together. Then try `tone(pin, frequency)` instead, with the frequency mapped from 200 to 2000, and decide which version sounds better.

New word	What it means
Potentiometer	A resistor with a slider, used as an adjustable knob.
Wiper	The middle leg of a pot, which picks up the changing voltage.
Voltage divider	A circuit that splits a voltage into a smaller share.
Decision ladder	A chain of if - else if - else, checked in order.